

Aun Siddiqui

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WORK EXPERIENCE

META - Software Engineer, Machine Learning March 2024 - Present

- **Generative Recommendations (Post-Training):** Currently developing post-training methods for a generative recommendation system using Group Relative Policy Optimization (GRPO) to align model outputs with downstream engagement objectives and improve recommendation quality.
- **VLM Evaluation Framework:** Architected a scalable evaluation framework benchmarking diffusion models using Visual Language Models (VLMs). Fine-tuned custom judge models to quantify high-level features (artifacts, motion, abnormalities, consistency, visual text correctness), directly informing ranking logic for the Meta AI app.
- **Multi-Modal RAG System:** Researched and deployed a Retrieval-Augmented Generation (RAG) system for intelligent soundtrack pairing. Developed a pipeline to synthesize video semantic embeddings and match them against music embeddings in a high-dimensional latent space.
- **Predictive Modeling & Optimization:** Developed a multi-task learning architecture to predict engagement and report risks for Facebook Groups. Achieved a **15% increase in post approvals** with neutral impact on safety reports.
- **Infrastructure Efficiency:** Refactored ML pipelines to exploit shared feature extraction across workflows. Reduced total compute costs by **10%** and implemented real-time monitoring to prevent feature drift during inference.

DEKA Research and Development - Machine Learning Engineer January 2021 - February 2024

- **Deep Learning for Mapping:** Designed, trained and productionalized an image segmentation model that successfully generated High-Definition Map from aerial imagery for an autonomous last-mile delivery robot, successfully reducing the total map creation time by ~44%.
- **Architectural Research:** Enhanced model performance by implementing **Self-Supervised Contrastive Learning** (based on *Li et al.*) for high-resolution imagery. Refined outputs using custom feature extraction backbones and advanced loss functions (Focal, Active Boundary) to mitigate class imbalance.
- **Active Learning Pipeline:** Led the development of a novel active learning loop leveraging uncertainty quantification to prioritize high-value annotation data. Resulted in a **20% performance improvement** while minimizing labeling costs.
- **Metric Learning:** Built a high-dimensional image search and deduplication system using learned embeddings, facilitating efficient querying across the training corpus.
- **Engineering Excellence:** Reduced technical debt by **30%** through modular refactoring and established rigorous documentation standards, cutting onboarding time by **50%**.

UHV Technologies - Software Engineer September 2020 - January 2021

- Developed a tool to identify roots in minirhizotron images, and quantify parameters (such as root length, and area/density at a certain depth) required by the biologists to gauge the health of plants.
- Developed a real-time object detection tool for identifying scrap metal pieces (~500ms) to maintain a steady flow on conveyor belts in the production pipeline. Subsequently classified these pieces by mineral composition using a machine learning model that integrated data from X-Ray Fluorescence and camera sensors.

Intelligent System and Robotics Lab - Research Assistant May 2020 - September 2021

- **Bayesian Deep Learning:** Designed an end-to-end multi-stream CNN to predict gaze vectors for Human-Robot Interaction (HRI). Implemented **Monte Carlo Dropout** to approximate Bayesian inference, enabling the quantification of predictive uncertainty.
- **Performance:** Achieved a gaze prediction accuracy of **10.2° ± 0.9°**, providing robust confidence intervals critical for safe robot interaction.

SKILLS

Programming Languages: Python, C++, C, HTML, SQL, Bash

Software and Tools: AWS, Airflow, CMake, Docker, Eigen, Gazebo, Git, Keras, Linux, MATLAB, ONNX, OpenCV, PCL, PyTorch, Robot Operating System, Selenium, Sci-kit learn, SolidWorks, QtCreator, TensorFlow, and TensorRT

EDUCATION

Colorado School of Mines May 2019

Master of Science in Mechanical Engineering, Specialization Robotics

University of Mumbai May 2016

Bachelor of Engineering in Mechanical Engineering